

torch.nanmedian#

<https://pytorch.org/docs/stable/generated/torch.nanmedian.html>

Description:

Returns the median of the values in input, ignoring NaN values. This function is identical to `torch.median()` when there are no NaN values in input. When input has one or more NaN values, `torch.median()` will always return NaN, while this function will return the median of the non-NaN elements in input. If all the elements in input are NaN it will also return NaN. input (Tensor) – the input tensor. Returns a namedtuple (values, indices) where values contains the median of each row of input in the dimension dim, ignoring NaN values, and indices contains the index of the median values found in the dimension dim.

Function Signature

```
torch.nanmedian(input) → Tensor#
```

Parameters

```
torch.nanmedian(input, dim=-1, keepdim=False, *, out=None)
```

Parameters

Keyword Arguments

Parameters

Keyword

out ((Tensor, Tensor), optional) – The first tensor will be populated with the median values and the second tensor, which must have dtype long, with their indices in the dimension dim of input.

Code Examples

```
>>> a = torch.tensor([1, float('nan'), 3, 2])
>>> a.median()
tensor(nan)
>>> a.nanmedian()
tensor(2.)
```

```
>>> a = torch.tensor([[2, 3, 1], [float('nan'), 1,
float('nan')]])
>>> a
tensor([[2., 3., 1.],
        [nan, 1., nan]])
>>> a.median(0)
torch.return_types.median(values=tensor([nan, 1., nan]),
indices=tensor([1, 1, 1]))
>>> a.nanmedian(0)
torch.return_types.nanmedian(values=tensor([2., 1., 1.]),
indices=tensor([0, 1, 0]))
```

Detailed Documentation

Documentation

Returns the median of the values in input, ignoring NaN values.

This function is identical to `torch.median()` when there are no NaN values in input. When input has one or more NaN values, `torch.median()` will always return NaN, while this

function will return the median of the non-NaN elements in input. If all the elements in input are NaN it will also return NaN.

input (Tensor) – the input tensor.

Returns a namedtuple (values, indices) where values contains the median of each row of input in the dimension dim, ignoring NaN values, and indices contains the index of the median values found in the dimension dim.

This function is identical to `torch.median()` when there are no NaN values in a reduced row. When a reduced row has one or more NaN values, `torch.median()` will always reduce it to NaN, while this function will reduce it to the median of the non-NaN elements. If all the elements in a reduced row are NaN then it will be reduced to NaN, too.

input (Tensor) – the input tensor.

dim (int, optional) – the dimension to reduce. If None, all dimensions are reduced.

keepdim (bool, optional) – whether the output tensor has dim retained or not. Default: False.

out ((Tensor, Tensor), optional) – The first tensor will be populated with the median values and the second tensor, which must have dtype long, with their indices in the dimension dim of input.